

BECU Framework

G-VAR MED:

A Stability-Oriented Multi-Modal Consistency Architecture

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1 Prologue

Modern clinical workflows rely on a combination of imaging, anatomical knowledge, mechanical interaction, and human expertise.

Despite technological progress, persistent challenges remain, including uncertainty in localization, tissue micro-movements, and discrepancies between pre-operative imaging and real-time tissue behavior.

The BECU (Biomechanical–Electromechanical Consistency Unit) framework introduces a stability-oriented approach in which clinical progression occurs only when all available information sources remain in agreement.

Rather than replacing human judgment, BECU acts as a supportive consistency layer across the imaging–diagnostic–interventional–therapeutic axis.

2 Intended Research Audience

This framework is intended for academic, engineering, and research-oriented discussion across fields involving:

- medical imaging,
- multimodal sensing,
- surgical navigation,
- deterministic control systems,
- biomedical engineering,
- real-time verification architectures,
- stability-oriented procedural systems.

The document is presented as a conceptual and architectural framework intended to support future research, simulation, validation, and interdisciplinary technical discussion.

It is not presented as a certified medical device, clinical protocol, or autonomous decision-making system.

3 Foundational Principle

The framework is based on a simple principle:

A clinical action should proceed only when independent information sources describing the same physical reality remain consistent.

BECU operates as a deterministic and traceable consistency architecture.

If agreement is maintained, the workflow may proceed.

If inconsistency appears, the system transitions into a stable, non-progressing state until agreement is restored.

4 Agreement Modalities

The framework evaluates four independent modalities:

- Geometric Agreement
- Mechanical Agreement
- Layer Agreement
- Dynamic Agreement

Each modality represents a different physical interpretation of the same anatomical reality.

4.1 Geometric Agreement

Geometric agreement evaluates alignment between the current position and the expected anatomical model derived from imaging and planning data.

Sources include:

- MRI geometry
- anatomical models
- planned trajectories
- segmentation boundaries

4.2 Mechanical Agreement

Mechanical agreement evaluates whether tissue interaction matches the expected mechanical profile.

Sources include:

- tissue resistance
- compliance patterns
- stiffness transitions
- micro-force signatures

4.3 Layer Agreement

Layer agreement evaluates whether anatomical layers appear in the expected sequence and depth.

Sources include:

- layer transitions
- micro-texture changes
- structural boundaries
- depth-dependent signatures

4.4 Dynamic Agreement

Dynamic agreement evaluates whether temporal tissue behavior remains consistent with the expected procedural environment.

Sources include:

- micro-movements
- physiological shifts
- tissue dynamics
- tool–tissue interaction behavior

5 Consistency Logic

The framework evaluates whether all modalities remain simultaneously consistent.

If agreement is maintained across modalities, progression may continue.

If any modality becomes inconsistent, the system transitions into a stable holding condition until verification is restored.

The framework is intentionally deterministic and conservative. It does not attempt to predict future states or replace operator judgment.

6 Safe-State Logic

A central principle of BECU is the concept of the safe state.

When inconsistency is detected, the system:

- pauses progression,
- maintains stability,
- requests confirmation,
- waits for restored agreement.

The safe state is not treated as an error condition. It is a controlled and stable operational state.

BECU does not reject actions. Instead, it simply avoids progression under inconsistent conditions.

7 Stability Envelope

The Stability Envelope defines the acceptable operational region for all modalities.

Within this region:

- progression is permitted,
- modality agreement is preserved,
- system behavior remains stable and traceable.



Figure 1: BECU safe-state and recovery behavior.

Outside the envelope:

- progression pauses,
- consistency is re-evaluated,
- the system remains stable until alignment returns.

The framework evaluates only the present state. It does not attempt to infer or predict future tissue behavior.

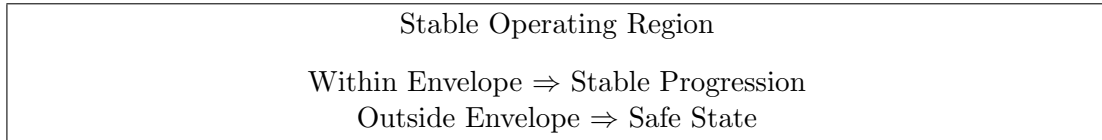


Figure 2: Conceptual representation of the Stability Envelope.

8 Unified Cross-Modal Stability Principle

A central characteristic of the BECU framework is that the system does not rely on a single imaging modality, isolated signal source, or fixed technological environment.

Modern clinical systems increasingly operate across heterogeneous imaging regimes, including conventional 1.5T MRI, high-resolution 3T systems, and emerging ultra-high-field 7T environments. While these platforms provide different levels of spatial resolution and signal sensitivity, they also introduce distinct forms of noise, distortion, instability, artifact amplification, and dynamic variability.

The BECU framework is designed to remain independent of any single field-strength assumption or isolated imaging interpretation.

Rather than treating one modality as absolute truth, BECU evaluates whether multiple independent representations of the same anatomical and procedural reality remain simultaneously consistent.

These representations include:

- geometric consistency,
- mechanical consistency,
- layer-based consistency,
- dynamic consistency.

Each modality provides a different projection of the same physical state.

The framework therefore operates as a cross-modal admissibility system in which progression is permitted only when the global multimodal state remains inside the Stability Envelope.

This principle allows the framework to remain stable even when individual modalities exhibit local uncertainty, temporary distortion, transient noise, or modality-specific variability.

The objective of BECU is not to eliminate noise, replace imaging systems, or infer hidden states through probabilistic prediction.

Instead, the framework continuously evaluates whether independent physical interpretations converge toward the same operational reality.

In this sense, the Stability Envelope acts as a modality-independent consistency geometry that anchors the clinical workflow to deterministic cross-modal agreement rather than dependence on a single signal source alone.

As imaging complexity, spatial resolution, and system sensitivity continue to increase across next-generation medical technologies, the importance of multimodal stability verification becomes progressively more critical.

The BECU framework therefore represents a stability-oriented architecture designed to preserve controlled progression across heterogeneous imaging and procedural environments through deterministic consistency evaluation.

9 Clinical Axis Integration

BECU operates across four clinical domains:

1. Imaging
2. Diagnostic
3. Interventional
4. Therapeutic

Each domain emphasizes different modalities while preserving the same stability-oriented logic.

9.1 Imaging Domain

The imaging domain focuses primarily on geometric consistency, including localization, anatomical boundaries, and planned trajectories.

9.2 Diagnostic Domain

The diagnostic domain emphasizes tissue characterization, layer transitions, and mechanical consistency.

9.3 Interventional Domain

The interventional domain continuously evaluates all modalities during procedural execution.

9.4 Therapeutic Domain

The therapeutic domain ensures that treatment is applied only under stable and verified conditions.

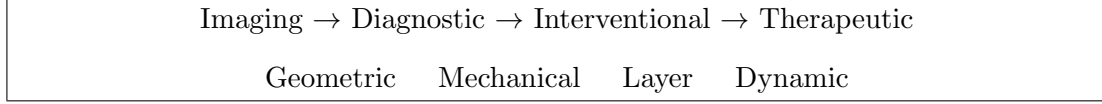


Figure 3: BECU operation across the clinical axis.

10 System Architecture

The BECU architecture is modular and consists of four primary layers:

1. Sensing Layer
2. Fusion Layer
3. Consistency Layer
4. Control Layer

10.1 Sensing Layer

Collects independent multimodal measurements from imaging, mechanical interaction, layer transitions, and tissue dynamics.

10.2 Fusion Layer

Normalizes and compares measurements against expected anatomical and procedural profiles.

10.3 Consistency Layer

Evaluates agreement across modalities and determines whether the system remains within stable operating conditions.

10.4 Control Layer

Controls progression, stabilization, and safe-state transitions.

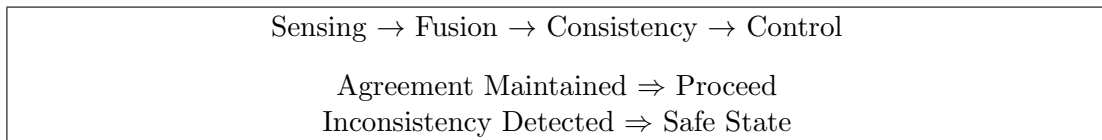


Figure 4: Closed-loop consistency evaluation process.

11 Real-Time Stability Operation

The framework operates as a continuous closed-loop system:

Sensing $\rightarrow$ Fusion $\rightarrow$ Consistency $\rightarrow$ Control

This loop ensures:

- continuous monitoring,
- deterministic behavior,
- immediate stabilization under inconsistency,
- traceable workflow progression.

The framework avoids prediction, probabilistic inference, and autonomous decision-making.

12 Transition Logic

Transitions between clinical domains occur only when the current state remains consistent.

If inconsistency appears during a transition:

- progression pauses,
- the current domain remains active,
- verification is requested,
- transition resumes only after restored agreement.

This ensures stable and predictable domain transitions.

13 Recovery Logic

Recovery occurs when all modalities return to stable agreement.

The recovery sequence includes:

1. re-evaluation,
2. restored consistency,
3. temporal verification,
4. controlled resumption.

The framework resumes progression only after stable agreement is continuously maintained.

14 Conclusion

The BECU framework introduces a deterministic, stability-oriented consistency architecture for multimodal clinical workflows.

Its central principle is simple:

Progression occurs only under verified agreement.

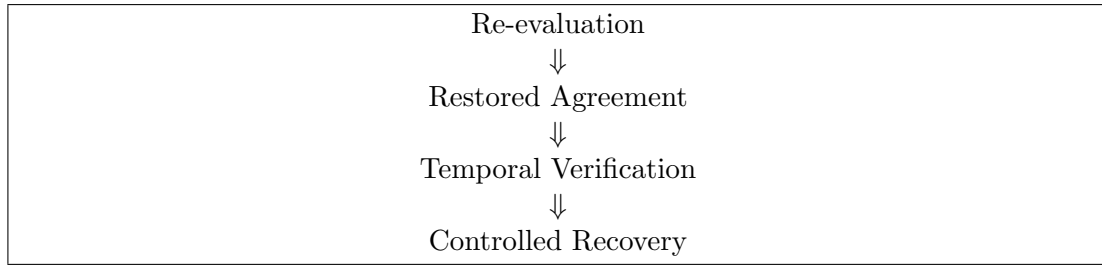


Figure 5: Recovery sequence after inconsistency resolution.

BECU does not replace operator judgment, prediction, or clinical expertise. Instead, it provides a structured consistency layer that supports stable, traceable, and controlled progression across the full clinical axis.

The framework emphasizes:

- multimodal verification,
- deterministic behavior,
- safe-state stabilization,
- transparency,
- real-time consistency evaluation.

In this way, BECU acts as a collaborative stability mechanism designed to support reliable and controlled clinical operation.

Disclaimer: This document presents a conceptual research framework intended for academic and engineering discussion. It does not represent a certified medical device, diagnostic platform, or clinical recommendation system.

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